Sawan Kumar

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Research Summary

I work at the intersection of **probabilistic machine learning, Gaussian processes, and scalable Bayesian inference** for scientific systems. My research develops **non-stationary kernel approximations, scalable operator-valued Gaussian processes, and diffusion-based variational inference** for expressive, calibrated uncertainty quantification.

Education

- **Indian Institute of Technology Delhi** — New Delhi, India
  PhD, Department of Applied Mechanics; GPA: 8.72/10.0 — *May 2022 – Present*
- **Delhi Technological University** — Delhi, India
  Bachelor of Technology, Mechanical Engineering; GPA: 8.50/10.0 — *Aug. 2016 – Aug. 2020*

Experience

- **Indian Institute of Technology Delhi** — New Delhi, India
  Teaching Assistant — *2022 – Present*
  - **Graduate ML Courses**: Instruction and student mentoring for *Probabilistic Machine Learning*, *Deep Generative Models*, and *Foundations of Machine Learning*.
- **Foundation for Smart Manufacturing (FSM), IIT Delhi** — New Delhi, India
  Research Intern — *Feb 2021 – Apr 2021*
  - **Robotics & Design**: Commissioned KUKA robotic arm; designed adaptive pallet fixtures to improve robotic cell efficiency.

Publications & Conferences

1. Kumar, S., & Chakraborty, S. (2026). Scalable Random Wavelet Features: Efficient Non-Stationary Kernel Approximation with Convergence Guarantees. *International Conference on Learning Representations.* **[ICLR]**
2. Kumar, S., Tripura, T., Nayek, R., & Chakraborty, S. (2026). From local interactions to global operators: Scalable Gaussian Process operator for physical systems. *Journal of Computational Physics, 114785.* **[JCP]**
3. Thakur, A., Kumar, S., Zahr, M., & Chakraborty, S. (2025). Scalable h-adaptive probabilistic solver for time-independent and time-dependent systems. **[CMAME]**
4. Kumar, S., Nayek, R., & Chakraborty, S. (2025). Towards Gaussian Process for Operator Learning: An Uncertainty-Aware Resolution Independent Operator Learning Algorithm. **[CMAME]**
5. Tushar, T., Kumar, S., & Chakraborty, S. (2025). Fusion-Based Constitutive Model (FuCe): Toward Model-Data Augmentation in Constitutive Modeling. **[IJMSD]**
6. Kumar, S., Nayek, R., & Chakraborty, S. (2024). Neural Operator Induced Gaussian Process Framework for Probabilistic Solution of Parametric PDEs. **[CMAME]**
7. Kumar, S., Nayek, R., & Chakraborty, S. (2023). Can an Operator Learning Framework Benefit from Gaussian Process? *9th International Congress on Computational Mechanics and Simulations.* [ICCMS]

Awards & Achievements

- **Best Oral Presentation – ICCMS 2025**: Awarded for the work on uncertainty-aware scalable probabilistic models.
- **GATE Qualified (ME & XE)**: Graduate Aptitude Test in Engineering, 2022
- **Junior Mathematics Olympiad**: Secured AIR 24
- **INSPIRE Scholar**: Department of Science & Technology, India

Research Areas & Tools

Methods: Bayesian deep learning, probabilistic ML, Gaussian processes, operator learning, generative models
Tools: PyTorch, JAX, Pyro, BoTorch, TensorFlow, Python, Julia, C++, MATLAB, Git, Docker, Linux, HPC
Key Contributions: Scalable GP Operators · Non-Stationary Kernel Approximation · Physics-aware Deep GPs